

Testing a Micro/Macro Dual-Scale Hypothesis

on $K3 \times T^2$:

Scale-Factor Duality, the Cohen–Kaplan–Nelson Bound,
and the Dark-Energy Length in Lean 4

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Abstract

We report `DualScaleCosmology` (Lean 4, toolchain v4.33.1, Mathlib), a companion library that formally tests one reading of a “micro/macro dual-scale” hypothesis motivating this research programme: that a microscopic length $\ell_{\text{micro}} \sim \ell_P$ and a macroscopic length $\ell_{\text{macro}} \sim H_0^{-1}$ are linked by non-perturbative duality. The hypothesis is not, and cannot be, a theorem; we instead formalize the two pieces of the literature that actually bear on it — string-cosmology scale-factor duality (Gasperini–Veneziano) and the Cohen–Kaplan–Nelson (CKN) ultraviolet/infrared bound — and test their numerical and structural consequences. The results are mixed and mostly negative for the literal hypothesis. Taken literally, ℓ_P as the ultraviolet cutoff at H_0^{-1} fails the CKN bound by more than 10^{30} (Tier A arithmetic on Tier L constants). Read instead as a T-dual pair, the two scales saturate the CKN bound exactly at their self-dual length $s = \sqrt{\ell_P c/H_0} \approx 47 \mu\text{m}$; we then prove, as an exact identity, that s equals the observed dark-energy length $\Lambda^{-1/4}$ up to a fixed, computable factor $(8\pi/3\Omega_\Lambda)^{1/4}$ — so the numerical coincidence often quoted between such lengths is algebra, not independent evidence. As a string (Regge) scale s is excluded by collider data by more than 10^{30} in α' ; as the radius of a single extra dimension it is disfavored only by an order-one factor, indistinguishable at this level of rigor from the “dark dimension” proposal of Montero–Vafa–Valenzuela. We also report, as a matter of methodological record, a citation error caught only by computing a numeric consequence of it, and an infrastructure misconfiguration that silently starved a local prover of its models for most of a work session. All Lean content is Tier A (44 declarations, 31 theorems, 0 `sorry`, standard axioms only); every physical identification is stated as Tier L (literature) or Tier C (this project’s own reading), never conflated with the former. We close with a concrete list of open problems and proposed research directions for the community.

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1 Introduction

1.1 The hypothesis under study

Perturbative superstring theories fix the fundamental string tension $T = 1/(2\pi\alpha')$ near the Planck scale, $\ell_s \sim \ell_P \sim 10^{-35}$ m. A dual-scale framework, motivating this programme’s earlier papers [1, 2], proposes instead a pair of dynamically coupled length scales: a microscopic scale $\ell_{\text{micro}} \sim \ell_P$ (quantum-gravity corrections, ultraviolet completion) and a macroscopic scale $\ell_{\text{macro}} \sim H_0^{-1}$ or an intermediate galactic/dark-sector scale, linked by non-perturbative duality transformations, scale-invariance breaking, or topological defects. This is *not* a theorem anywhere in the literature, and this paper does not treat it as one. It is a Tier C proposal, and the question this paper answers is narrower and more disciplined: which pieces of existing physics literature actually relate a micro and a macro length by a precise mathematical statement, and what do their kernel-checked consequences say about the hypothesis as stated?

1.2 Two literature anchors, not a re-derivation

Two, and — after a systematic search — only two, published results were found that relate a Planck-scale ultraviolet input to a cosmological infrared scale with a precise statement rather than a general expectation.

1. **Scale-factor duality.** Gasperini and Veneziano [4] show that the scale factor $a(t)$ of an FRW background is odd under $a \mapsto a^{-1}$ in exactly the $O(d, d)$ sense that a compactification radius is odd under Buscher T-duality $R \mapsto \alpha'/R$ (paper 1’s Theorem 2.1). This is a genuine duality, but it is a symmetry of the early-universe string effective action, not on its own a statement relating today’s Hubble scale to the Planck scale.
2. **The Cohen–Kaplan–Nelson bound.** Cohen, Kaplan, and Nelson [5] show that an effective field theory with ultraviolet cutoff Λ in a box of infrared size L , demanding no state already collapsed to a black hole, satisfies $L^3\Lambda^4 \lesssim LM_P^2$. This is an actual inequality — not a duality — tying a micro (UV) scale to a macro (IR) scale, and the authors themselves instantiate it with L at the current cosmological horizon. If any published result deserves the name “micro/macro dual scale,” this is it, and it predates this project’s hypothesis by 28 years.

Nothing else in the literature search (cosmic-string tension bounds, the swampland distance conjecture, the “dark dimension” proposal) relates a Planck-scale and a Hubble-scale length by a precise, checkable statement rather than an order-of-magnitude expectation; those sources are used below, but as tests of the two anchors above, not as further anchors.

1.3 Contents and how to read the tier labels

Section 2 restates this project’s epistemic tier system. Section 3 formalizes scale-factor duality. Section 4 formalizes the CKN bound, including a citation error caught and corrected in the course of this work. Section 5 tests the literal hypothesis numerically and finds it fails. Section 6 finds where the two anchors meet: a self-dual length at which the CKN bound is saturated. Section 7 gives that length two physical readings — excluded as a string scale, disfavored (not excluded) as an extra-dimension radius — and Section 8 proves it is, as an exact identity, the observed dark-energy length. Section 9 formalizes cosmic F-strings, the one genuinely micro object that can be cosmologically long. Section 10 tests whether the swampland distance conjecture is an independent third route, or shares scale-factor duality’s structure. Section 11 records what was deliberately *not* formalized, and why. Section 12 summarizes what is and is not proved. Section 13 reports two verification-discipline case studies. Section 14 and Section 15

close with open problems and proposed research directions for the community. Appendix A is a reproducibility appendix.

2 Epistemic tiers

As in this programme’s earlier papers, every claim below carries one of three tags.

- **Tier A — Lean kernel** — accepted by the Lean 4 kernel: `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `admit`, no `native_decide`, no project-declared axiom.
- **Tier L — literature** — established in the cited literature, quoted with a file and line range in `papers/foundations/*.txt` that a reader can `grep`, not re-derived here.
- **Tier C — conjecture** — a modeling choice or conjecture introduced by this project. **The identification of a Lean object with a physical quantity is never Tier A**, no matter how the surrounding prose reads.

Two conventions specific to this paper, both forced by incidents recorded in Section 13: every numeric bound quoted from a source is rounded in the direction that *weakens* the conclusion being drawn (never in the direction that helps it); and every Lean theorem name is checked to be plain ASCII, because this project’s own audit tooling silently mis-parses non-ASCII identifiers (Section 13).

3 Scale-factor duality

Gasperini–Veneziano [4] (ll. 372–373): “the Hubble parameter is odd under scale factor duality, $H \rightarrow -H$, so that a duality transformation $a \rightarrow a^{-1}$ maps an expanding solution” to a contracting one; the self-dual case is “a metric that satisfies $a^{-1}(t) = a(-t)$ ” (ll. 639–640). **Tier L — literature**.

Theorem 3.1 (Scale-factor duality is an involution). **Tier A — Lean kernel** For $a \neq 0$, $(\text{ScaleFactorDuality.scaleFactorDual_inv}) (a^{-1})^{-1} = a$.

Theorem 3.2 (Self-dual point). **Tier A — Lean kernel** For $a > 0$, $a^{-1} = a \iff a = 1$ ($\text{scaleFactorDual_fixed_iff}$) — the $t = 0$ instance of $a^{-1}(t) = a(-t)$.

Theorem 3.3 (The Hubble parameter is odd under the duality). **Tier A — Lean kernel** For $a : \mathbb{R} \rightarrow \mathbb{R}$ differentiable at t with $a(t) \neq 0$, with $H(t) := a'(t)/a(t)$,

$$H(s \mapsto a(s)^{-1})(t) = -H(a)(t) \quad (\text{hubble_dual}).$$

This is a genuine differentiation, via Mathlib’s `HasDerivAt.inv`, not an arithmetic shadow. An earlier draft of this work left Theorem 3.3 as Tier L, on the assumption that real differentiation was out of scope for this project’s usual style; that assumption was checked against confirmed proof work in a sibling repository and found wrong (Section 13).

Theorem 3.4 (Transferred dual-scale bound). **Tier A — Lean kernel** For $a > 0$, $a + a^{-1} \geq 2$ ($\text{cosmoDualScale_ge_two}$), the exact algebraic transfer of Stream 2’s torus bound $\text{tr } G + \text{tr } G^{-1} \geq 2d$ [3] to the scale factor.

4 The Cohen–Kaplan–Nelson bound

Case study: a citation error caught only by computing something

An earlier commit of this work stated CKN’s eq. (2) as $L^3\Lambda^4 \leq M_P^2$. This compiled as valid Lean and was reported, correctly, as Tier A — but it is *dimensionally inconsistent* ($[L^3\Lambda^4] = \text{GeV}, [M_P^2] = \text{GeV}^2$) and therefore not what the paper says. The kernel cannot catch a false Tier L citation; only someone re-reading the source can. The error was found not by re-reading in the abstract, but by trying to *use* the statement for the numeric test of Section 5 and finding the units did not close. Re-extracting the source with `pdftotext -layout` (rather than trusting the first, unlayouted pass) gives the actual eq. (2): $L^3\Lambda^4 \lesssim LM_P^2$ (ll. 77–78) [5], dimensionally consistent, and — dividing by $L > 0$ — exactly the well-known holographic-dark-energy scaling $L^2\Lambda^4 \lesssim M_P^2$. Fixing the exponent *simplified* the Lean proofs below: no fractional-power (`Real.rpow`) machinery is needed for the corrected statement, only one `nlinarith` step.

CKN, ll. 77–78, 80 [5]: $L^3\Lambda^4 \lesssim LM_P^2$, so “the IR cutoff scales like Λ^{-2} .” Tier L — literature.

Theorem 4.1 (CKN bound, simplified form). Tier A — Lean kernel For $L > 0$, $L^3\Lambda^4 \leq LM^2 \implies L^2\Lambda^4 \leq M^2$ (`CKNBound.ckn_bound`).

Theorem 4.2 (Solved for the combination $L\Lambda^2$). Tier A — Lean kernel For $L, \Lambda, M > 0$, $L^2\Lambda^4 \leq M^2 \implies L\Lambda^2 \leq M$ (`ckn_L_Lambda_sq_le`): since $L^2\Lambda^4 = (L\Lambda^2)^2$, this is a one-step square-root extraction, not a fractional power.

Corollary 4.3 (Solved for each variable). Tier A — Lean kernel $L \leq M/\Lambda^2$ (`ckn_L_le`) and $\Lambda^2 \leq M/L$ (`ckn_Lambda_sq_le`).

5 Testing the literal hypothesis

CKN’s own worked example, ll. 113–114 [5]: fixing the IR cutoff L at the current cosmological horizon, eq. (2) implies $\Lambda \sim 10^{-2.5} \text{ eV}$. Tier L — literature. We use this number directly, rather than re-deriving H_0^{-1} in Planck units from CODATA constants, to avoid re-introducing exactly the kind of unit-conversion error of the case study above.

Theorem 5.1 (The literal hypothesis fails by $\geq 10^{30}$). Tier A — Lean kernel With $\Lambda_{\text{CKN}} := 3.2 \times 10^{-3} \text{ eV}$ (CKN’s value, rounded up, conservative) and $M_P := 1.2 \times 10^{28} \text{ eV}$ (CODATA 2018 Planck energy, rounded down, conservative),

$$M_P \geq 10^{30} \Lambda_{\text{CKN}} \quad (\text{CKNInstance.planckEnergy_gt_ckn_horizon_cutoff}).$$

The true ratio is $\approx 3.9 \times 10^{30}$; both roundings understate it. **Reading:** if the Stream 3 hypothesis’s “ $\ell_{\text{micro}} \sim \ell_P$ ” is the ultraviolet cutoff the CKN bound would allow at “ $\ell_{\text{macro}} \sim H_0^{-1}$,” it is not — by 30 orders of magnitude. This is a direct, literature-sourced counter-example to the literal reading of the hypothesis.

6 Where the bound meets the duality: the self-dual length

Theorem 5.1 tests one specific reading. This section asks what *does* follow if ℓ_{micro} and ℓ_{macro} are instead read as a T-dual pair — the reading Section 3 actually formalizes — with $\alpha' := \ell_{\text{micro}}\ell_{\text{macro}}$ and CKN’s Planck mass identified with $M = 1/\ell_{\text{micro}}$ (natural units). Both identifications are this project’s choices, stated here as explicit hypotheses, not results. Tier C — conjecture.

Theorem 6.1 (The pair is a scale-factor-duality pair). Tier A — Lean kernel In units of $s := \sqrt{\ell_{\text{micro}} \ell_{\text{macro}}}$, $\ell_{\text{macro}}/s = (\ell_{\text{micro}}/s)^{-1}$ (`SelfDualCutoff.selfDual_pair_is_scaleFactorDual`).

Theorem 6.2 ($\text{CKN} \iff \text{UV length at least the self-dual length}$). Tier A — Lean kernel With $L = \ell_{\text{macro}}$, $M = 1/\ell_{\text{micro}}$, $\Lambda = 1/\ell_{\text{UV}}$,

$$L^2 \Lambda^4 \leq M^2 \iff \ell_{\text{micro}} \ell_{\text{macro}} \leq \ell_{\text{UV}}^2 \quad (\text{ckn_iff_uvLength_ge_selfDual}),$$

saturated at equality (`ckn_saturated_at_selfDual`). Being an iff between non-equivalent-looking statements, this cannot be vacuous: it fails in each direction for suitable ℓ_{UV} .

Proposition 6.3 (Numeric bracket). Tier A — Lean kernel With $\ell_P = 1.616255 \times 10^{-35} \text{ m}$ (CO-DATA 2018) and $c/H_0 = 1.3672 \times 10^{26} \text{ m}$ ($H_0 = 67.66 \text{ km/s/Mpc}$, Planck 2018 VI [15], Table 2, via `astropy.cosmology.Planck18`),

$$40 \mu\text{m} \leq \sqrt{\ell_P c/H_0} \leq 70 \mu\text{m} \quad (\text{selfDual_length_sq_bracket}),$$

and CKN’s own horizon-scale cutoff length $\hbar c/\Lambda_{\text{CKN}}$ lies in the same bracket (`cknHorizon\_length\_bracket`): $s \approx 47.0 \mu\text{m}$, $\hbar c/\Lambda_{\text{CKN}} \approx 62 \mu\text{m}$.

Reading: the hypothesis’s “ $\ell_{\text{micro}} \sim \ell_P$ ” is not where the CKN bound’s implied micro cutoff sits (Theorem 5.1); the geometric mean $s = \sqrt{\ell_P \ell_{\text{macro}}}$ is.

7 Two physical readings of the self-dual length

7.1 As a string scale: excluded

If $\alpha' = s^2$ is the fundamental string’s Regge slope, then, since $M^2 = (N-1)/\alpha'$ at level N [14] (§1.3), string excitations sit near $\hbar c/s \approx 4 \text{ meV}$. CMS-EXO-19-012, 137 fb $^{-1}$ at 13 TeV, ll. 1044–1048 [9]: string resonances (Regge excitations of quarks and gluons, l. 63) are excluded below 7.9 TeV at 95% CL, in the low-string-scale models of its own references. Tier L — literature.

Theorem 7.1 (Excluded as a Regge slope). Tier A — Lean kernel $\ell_P c/H_0 \geq 10^{30} (\hbar c/7.9 \text{ TeV})^2$ (both sides m 2 ; `CosmicString.selfDual_alphaPrime_exceeds_cms_ceiling`).

This excludes only the *Regge-slope* reading; it says nothing about s as a *compactification radius* (Section 7.2), and it is model-dependent on CMS’s own low-string-scale models, stated as such.

7.2 As an extra-dimension radius: disfavored, not excluded

Torsion-balance tests (Lee et al., ll. 285–289 [10]): gravitational-strength Yukawa interactions are excluded for range $< 38.6 \mu\text{m}$; a single large extra dimension’s toroidal radius must be $< 30 \mu\text{m}$ at 95% CL. Tier L — literature.

Theorem 7.2 (Above the torsion-balance bound). Tier A — Lean kernel $(1.5 \times 30 \mu\text{m})^2 \leq \ell_P c/H_0$ (`DarkEnergyScale.selfDual_above_torsion_radius_bound`).

So identifying s directly with a large extra-dimension radius exceeds the experimental bound by a factor of only ≈ 1.6 — an order-one margin, not the 10^{30} of Theorem 7.1. This paper does *not* claim the extra-dimension reading is excluded.

8 The self-dual length is the dark-energy length

With $\rho_\Lambda = \Omega_\Lambda \cdot 3H_0^2/(8\pi G)$ (critical-density form used by Copeland–Kibble, eq. (4.1), ll. 430–434 [12]) and $G = \ell_P^2$ ($\hbar = c = 1$):

Theorem 8.1 (An exact identity). Tier A — Lean kernel

$$\rho_\Lambda^{-1} = \frac{8\pi}{3\Omega_\Lambda} (\ell_P H_0^{-1})^2 \quad (\text{DarkEnergyScale.rhoLambda_inv_eq}),$$

i.e. $\rho_\Lambda^{-1/4} = (8\pi/3\Omega_\Lambda)^{1/4} s$: the self-dual length is the dark-energy length $\Lambda_{\text{cc}}^{-1/4}$, up to a fixed, ℓ_P - and H_0 -independent number. Equivalently, in Montero–Vafa–Valenzuela’s (MVV) parameterization $l = \lambda \Lambda_{\text{cc}}^{-1/4}$ (l. 67 [11]), s has $\lambda_s = (3\Omega_\Lambda/8\pi)^{1/4}$, a pure number (*selfDual_lambda_eq*).

Why this blocks a natural-looking argument

MVV quote $\Lambda_{\text{cc}}^{-1/4} \approx 88 \mu\text{m}$ (l. 303). CKN’s own horizon cutoff length is $\approx 62 \mu\text{m}$ (Proposition 6.3), and $s \approx 47.0 \mu\text{m}$. *These are not three independent numbers agreeing:* by Theorem 8.1 they are the same formula evaluated with different order-one conventions (reduced vs. non-reduced Planck mass gives a factor $(8\pi)^{1/4} \approx 2.2$ in length). A numeric bracket test alone (*darkEnergyLength4_bracket*, giving $85\text{--}90 \mu\text{m}$ for $\Lambda_{\text{cc}}^{-1/4}$) would sit *outside* the earlier $[40, 70] \mu\text{m}$ bracket of Proposition 6.3 — which by itself is evidence the bracket is convention-dependent, not physical. No document in this programme should cite the numerical closeness of s , the CKN horizon length, and $\Lambda_{\text{cc}}^{-1/4}$ as independent corroboration.

Proposition 8.2 (λ_s vs. MVV’s range). Tier A — Lean kernel $\lambda_s^4 \in [0.5^4, 1)$, i.e. $\lambda_s \approx 0.535$ (*selfDual_lambda_above_mv_range*): inside MVV’s broad range $10^{-4} < \lambda < 1$ (l. 424) but $\geq 5 \times$ their central estimate’s upper end $\lambda \sim 10^{-1}$ (ll. 424–427). MVV separately exclude a light string tower at this scale, on different grounds (local EFT breakdown, ll. 313–316) — consistent with Theorem 7.1.

9 Cosmic F-strings: a genuine micro object at macro scale

A cosmic superstring is literally a micro object (string tension) stretched to cosmological length — the most direct candidate for the hypothesis’s mechanism. Copeland–Myers–Polchinski (CMP) [7] give the ten-dimensional (p, q) tension, eq. (3.3), ll. 502–506, and the four-dimensional tension $\mu = e^{2A} \bar{\mu}$ with warp factor e^{2A} , eq. (3.2), l. 497. Tier L — literature. For the fundamental string, $(p, q) = (1, 0)$, unwarped ($e^{2A} = 1$, a Tier C simplification that can only *lower* μ if relaxed), $G\mu/c^2 = \ell_P^2/(2\pi\alpha')$.

Theorem 9.1 (An observational bound on $G\mu$ is a lower bound on α'). Tier A — Lean kernel For $\alpha' > 0$, $G\mu/c^2 \leq g \iff \ell_P^2 \leq 2\pi g \alpha'$ (*CosmicString.fStringGmu_le_iff*). For the self-dual pair, $\alpha' = \ell_P L \implies G\mu/c^2 = \ell_P/(2\pi L)$ (*fStringGmu_selfDual*).

Proposition 9.2 (The self-dual α' gives a badly under-tense string). Tier A — Lean kernel $G\mu/c^2 \leq 10^{-61}$ for the self-dual pair (*selfDual_Gmu_below_cmp_window*), more than 50 orders below the lower end 10^{-11} of CMP’s brane-inflation window (eq. (5.1), ll. 782–783), whose own caveat (ll. 809–811) notes the window assumes inflaton-driven perturbations.

Not formalized: the defect-formation and horizon-scaling mechanism (Kibble, network scaling) that would let a cosmic string carry the macro scale in the first place — Section 11.

10 Dual towers and the swampland distance conjecture

The Swampland Distance Conjecture (SDC, due to Ooguri–Vafa, quoted at ll. 2167–2181 [6]): from any point P in moduli space there is a point Q at infinite geodesic distance and an infinite tower of states with $M(Q) \sim M(P)e^{-\alpha d(P,Q)}$, $\alpha > 0$. Tier L — literature.

Why eq. (2.100) itself is not formalized

The SDC’s own equation is asymptotic, with an unspecified α and an “ $\sim$ ”. Any Lean theorem about the bare expression $M_0 e^{-\alpha d}$ (positive, monotone, tends to zero) is a fact about `Real.exp`, not physics, and would restate its own definition — the exact vacuous-check failure a sibling project in this ecosystem retracted after finding it in its own work (see Section 13). We formalize instead Palti’s *motivating remark* just above the SDC box (ll. 2158–2161): dual towers whose mass product is constant, under an explicit Tier C model $M_1 M_2 = M_0^2$.

Theorem 10.1 (The dual-tower structure is P3.1’s involution). Tier A — Lean kernel Under $M_1 M_2 = M_0^2$ (Tier C), $M_1 + M_2 \geq 2M_0$ (`DualTower.dualTower_sum_ge`), proved by rescaling to $a = M_1/M_0$ and invoking Theorem 3.4 unchanged. So the dual-tower structure is not an independent piece of algebra: it is scale-factor duality again.

Proposition 10.2 (Negative controls). Tier A — Lean kernel Without $M_1 M_2 = M_0^2$, both $\forall M_0, M_1, M_2 > 0, 2M_0 \leq M_1 + M_2$ and $\forall M_0 > 0, \min(M_1, M_2) \leq M_0$ are false, by explicit kernel-checked counterexample (`dualTower_sum_ge_needs_product`, `dualTower_min_le_needs_product`). Theorem 10.1 is therefore not vacuous.

Proposition 10.3 (Bridge to the exponential form). Tier A — Lean kernel $(M_0 e^{-\alpha d})(M_0 e^{\alpha d}) = M_0^2$ (`expTowerPair_product`) — a fact about `Real.exp`, connecting the SDC’s shape to the model above, not a formalization of the SDC itself.

11 Closed as out of scope: defect networks and domain walls

How a defect network grows to carry the macro scale is a *simulation* result, not a theorem: string networks reach a scaling regime $\xi/t \approx \text{const}$ by “general agreement from several simulations” (Copeland–Kibble, ll. 421–423 [12]), with $\rho_{\text{str}}/\rho = 8\pi G\mu\gamma^2/(3\nu^2)$ (eq. (4.1)); domain walls reach $\rho_{\text{wall}} = A\sigma/t$ with $A \approx 0.8 \pm 0.1$ from field-theory simulations (Saikawa, eqs. (2.17)–(2.18), ll. 269–283 [13]). Tier L — literature. Formalizing $\rho_{\text{str}} = \mu/\xi^2$ or $\rho_{\text{wall}} = A\sigma/t$ in Lean would only restate their own defining equations — exactly the vacuity failure this project’s own gate exists to catch (Section 13) — so neither is formalized. The one structural fact recorded, Tier L, not proved: a defect network carries the macro scale *by causality* ($\xi \lesssim t$ [12], ll. 272–276), while the micro scale enters only through the tension; the two are not related by a duality in this mechanism, unlike Section 3.

12 What is, and is not, proved

Scope of Lean 4 verification

Is proved (Tier A, kernel-checked, 44 declarations / 31 theorems, 0 sorry, standard axioms only): that scale-factor duality is an involution with self-dual point $a = 1$ and odd Hubble parameter; that the corrected CKN bound $L^2\Lambda^4 \leq M^2$ follows from CKN’s own eq. (2) and implies $L\Lambda^2 \leq M$; the numeric gap $\geq 10^{30}$ between CKN’s own horizon cutoff and the Planck energy; that the CKN bound holds *iff* the UV length is at least a T-dual pair’s self-dual length, with that length bracketed at $40\text{--}70\,\mu\text{m}$; that this length is excluded as a Regge slope by $\geq 10^{30}$ in α' and lies only $\approx 1.6\times$ above a torsion-balance bound as an extra-dimension radius; the *exact identity* between that length and the dark-energy length; a cosmic F-string’s tension formula and its numeric position 50+ orders below a literature brane-inflation window; that the dual-tower structure behind the swampland distance conjecture is scale-factor duality’s own involution, with kernel-checked negative controls.

Is not proved, anywhere in this paper: the micro/macro dual-scale hypothesis itself, in any reading; that any of the Tier C identifications used above (T-dual pair, $M = 1/\ell_{\text{micro}}$, $M_1 M_2 = M_0^2$) is physics rather than a modeling choice; the SDC’s own asymptotic statement, eq. (2.100); the defect-network scaling mechanism or domain-wall dynamics (Section 11); that a string scale of $\sim 50\,\mu\text{m}$ or an extra dimension of that size exists; and, as always in this project, that any Lean object *is* the physical system it is named after.

13 Methods: two verification-discipline case studies

Beyond the citation-transcription error of Section 4, two further incidents are recorded here because they generalize beyond this paper.

13.1 A false “models still resident” claim

An early version of this work’s workflow log stated that both local prover models were confirmed resident on the GPU after an Ollama restart. This was false: the restart had killed the system service and launched a user-level server pointing at an empty model directory, which then crash-looped on a port conflict for roughly two hours; the `/api/tags` check that appeared to confirm the models was run *before* the restart, not after. No result in this paper depends on the local prover having been available during that window, because every Stream 3 goal through Section 10 was closed by the orchestrating model directly (the orchestrator in this programme is a model working under the author’s direction, not a human checking each step). The error was found only when a later prover invocation returned HTTP 404 for every attempt in four seconds — an implausibly fast failure for a model that should take over a minute to cold-load. **Lesson:** a status check taken before an action is not evidence about the state after it, and a suspiciously fast “success” is itself a signal to distrust.

13.2 Identifier ASCII-only

An early draft named a theorem `ckn_Λ_1e`. This project’s own axiom-audit tool matches declaration names with the regular expression `[A-Za-z0-9_'.]+`, which does not include Greek letters; it silently truncated the name to `ckn_` and then reported that no such declaration existed, rather than reporting a parse failure. No existing declaration anywhere in this repository’s seven prior libraries used a non-ASCII identifier (checked by `grep` before this rule was written); this project now keeps that invariant explicitly.

Both incidents, together with the citation error of Section 4, are why this paper reports their discovery rather than only their fixes: the discipline that catches an error is at least as reportable as the theorem it protects.

14 Open problems for the community

Each item below names the specific gap, not just the headline claim, following this project’s own convention that a precise open problem is worth more than a vague one.

1. **The extra-dimension reading is untested at the right precision.** Theorem 7.2 shows only an order-one margin against a 2020 torsion-balance bound. A tighter, more recent bound (or the forecast sensitivity of a proposed experiment) formalized the same way as Theorem 7.1 would either exclude or further constrain the reading, resolving the current “disfavored, not excluded” verdict one way or the other.
2. **The Tier C identifications are never derived.** $\alpha' = \ell_{\text{micro}}\ell_{\text{macro}}$ and $M = 1/\ell_{\text{micro}}$ (Section 6) and $M_1M_2 = M_0^2$ (Section 10) are this project’s modeling choices, stated as hypotheses. A derivation of any of them from an explicit worldsheet or effective-field-theory argument — rather than an assumption — would upgrade the corresponding results from conditional Tier A to something closer to Tier L+A.
3. **The reduced-vs-non-reduced Planck-mass ambiguity is prose, not Lean.** The factor $(8\pi)^{1/4} \approx 2.2$ separating conventions in Theorem 8.1 is discussed but not formalized as an explicit, named parameter; a Lean statement carrying the convention as an argument (rather than a fixed numeral) would make every downstream numeric claim convention-parametric and auditable.
4. **The SDC itself remains unformalized, for a principled reason (Section 10).** An explicit, non-asymptotic toy moduli space — for instance the circle compactification already present in Stream 1 [2], where geodesic distance and tower mass are both concrete — could support a genuinely non-vacuous, non-asymptotic instance of eq. (2.100), where the general statement cannot.
5. **The defect-network mechanism (Section 11) has no non-vacuous target identified.** Saikawa’s wall-domination time $t_{\text{dom}} = 3M_{\text{Pl}}^2/(4A\sigma)$ (eq. (2.19)) is a genuine arithmetic consequence, not a restatement of the scaling law itself, and is a candidate for a first Tier A statement in this area once a specific σ is pinned to a source.
6. **Numeric brackets are hand-rounded, not interval arithmetic.** Every bracket in Sections 6 and 8 uses conservatively rounded decimal literals chosen by the authors. A Lean interval-arithmetic treatment computed directly from the CODATA/Planck-18 constants would remove the rounding-direction judgment call entirely and tighten every bracket.
7. **Relevance to $K3 \times T^2$ is stated, not derived.** If the “dark dimension” reading of s is taken seriously, the only natural single large direction on this programme’s own compactification is one T^2 circle, whose T-dual partner would then be sub-Planckian; nothing in this repository derives such a decompactification limit or checks it against Stream 2’s torus bound [3].

15 Proposed research directions

1. **A general “duality $\iff$ bound-saturation” lemma.** Theorem 6.2’s pattern — a monotone bound holds exactly when a length exceeds the self-dual point of a paired

duality — did not require anything specific to CKN beyond monotonicity in each variable. Formalizing this as an abstract lemma over an arbitrary duality-bound pair could turn this paper’s central observation into a reusable tool, applicable to other swampland-type UV/IR bounds (the species scale, the Weak Gravity Conjecture) with the same methodology.

2. **A systematic library of swampland bounds**, each tested the way Sections 5 and 7 test CKN: state the bound from its source, instantiate it at project-relevant scales, and report the numeric gap (or lack of one) as a Tier A fact. This would make “does hypothesis X survive bound Y” a repeatable, auditable procedure rather than a one-off calculation.
3. **Machine-generated negative controls as a general pattern.** Proposition 10.2 was written by hand; a tactic or script that, given a Tier A theorem with a nontrivial hypothesis, automatically searches for a counterexample when the hypothesis is dropped would let every future Stream check its own theorems for the vacuity failure discussed in Section 10 without a human writing the counterexample each time.
4. **Confronting the extra-dimension reading with forecast, not only current, data.** Item 1 of Section 14 generalizes: a Lean statement parameterized by an experimental sensitivity (rather than a fixed 2020 bound) could be re-evaluated as new torsion-balance or collider results appear, turning Theorem 7.2 into a living, re-checkable claim rather than a snapshot.
5. **The compile-time infrastructure recommended for this repository’s book (papers/book/chapters/ch38_open.tex, items 23–24)** — attribute-level Tier tracking and an anti-vacuity linter — would apply directly to future Stream 3-style work, where every new theorem in this paper had to be checked by hand against the vacuity failure of Section 10.
6. **A first continuous (non-arithmetic-shadow) theorem**, as recommended by an external review of this programme’s book: the classical equivalence of the Nambu–Goto and Polyakov actions, eliminating the worldsheet metric through its own equation of motion [14] (§1.3, ll. 1255–1280). The pointwise, finite-dimensional step is provable with Mathlib as it stands; the field-level calculus-of-variations statement is not yet, and closing that gap would be this programme’s first genuinely continuous physics theorem rather than an arithmetic or algebraic shadow of one.

16 Conclusion

Scale-factor duality and the Cohen–Kaplan–Nelson bound are formalized from pinned sources, and the dual-tower structure behind the swampland distance conjecture is shown to share scale-factor duality’s algebra rather than adding an independent route. Read literally, the micro/macro dual-scale hypothesis motivating this stream fails the CKN test by at least 10^{30} . Read as a T-dual pair, the two scales meet the CKN bound exactly at their self-dual length, and that length is — as an exact identity, not a numerical coincidence — the observed dark-energy length. It cannot be the fundamental string’s scale, and as an extra-dimension radius it is at present indistinguishable from an existing, unrelated proposal in the literature. None of this is a proof or a disproof of the hypothesis; it is a record of what two decades-old pieces of physics actually imply for it, kernel-checked, with every physical identification labelled for what it is. We consider the value of this exercise to lie exactly there — and in the two verification-discipline failures of Section 13, each caught by trying to use a claim rather than by reading it, as much as in the theorems themselves.

A Reproducibility

A.1 Build

```
lake build DualScaleStream2 StringTheoryFormalization StringTheoryFoundation \
DualScaleM24Formalization DoubleFieldTheory DualScaleValidation Lean5Corpus DualScaleCosmology
```

3781 jobs, 0 errors (release tag v3.4.0/v3.5.x). The job count is dominated by Mathlib’s own dependency graph, not this library’s proof content.

A.2 Axiom audit

```
python3 tools/axiom_audit.py DualScaleCosmology
```

reports **31 theorems**, **0 failing**, every declaration’s axiom footprint limited to `propext`, `Classical.choice`, `Quot.sound`. Across all eight libraries in this repository: **457 theorems**, **0 failing** (release v3.6.0, which added one theorem to Stream 2).

A.3 Statement lock

```
python3 tools/statement_lock.py -check $(find DualScaleCosmology -name '*.lean')
```

locks all 44 declarations across the 7 files of DualScaleCosmology: `ScaleFactorDuality.lean`, `CKNBound.lean`, `CKNInstance.lean`, `DualTower.lean`, `SelfDualCutoff.lean`, `CosmicString.lean`, `DarkEnergyScale.lean`.

A.4 Roadmap coverage

`docs/STREAM3_WORKFLOW.md` §6 records every item P3.1–P3.9 as either ✓ (closed with the theorems reported above) or explicitly closed as out of scope with the reason stated in Section 11, per the definition of done fixed before the last work package.

A.5 Sources pinned

The following are each hash-pinned in `papers/foundations/MANIFEST.md`, with the line range quoted at the point of use above: `hep-th/9211021`, `hep-th/9803132`, `1903.06239`, `hep-th/0312067`, `1303.5085`, `1911.03947`, `2002.11761`, `2205.12293`, `0911.1345`, `1703.02576`. CODATA 2018 and Planck 2018 VI constants via `astropy`.

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